PROJECT SCOPE AUTOMATION SYSTEM

Complete System Design, Solution Options & Step-by-Step Implementation Guide

Action Builders

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1. Executive Summary

Hubbard Builders currently prepares project scope documents manually. Project managers must open multiple Excel files, locate the relevant cost centres, copy line items, and paste them into a scope template. This process is time-consuming, error-prone, and inconsistent across projects.

This document presents a complete plan to replace that manual process with a digital, tablet-friendly system. Project managers will use a simple app or web form to select their project details, choose from 43 pre-loaded cost centres, pick the relevant line items, and automatically generate a formatted scope document — all in minutes, not hours.

Problem: Manual scope preparation from 43 cost centres takes significant time and risks errors.

Solution: A digital selection tool + automated document generator, built within your existing Microsoft 365 environment.

Recommended Approach: Power Apps + Power Automate + SharePoint — no external software, no extra licensing required.

2. Current Process & Problems

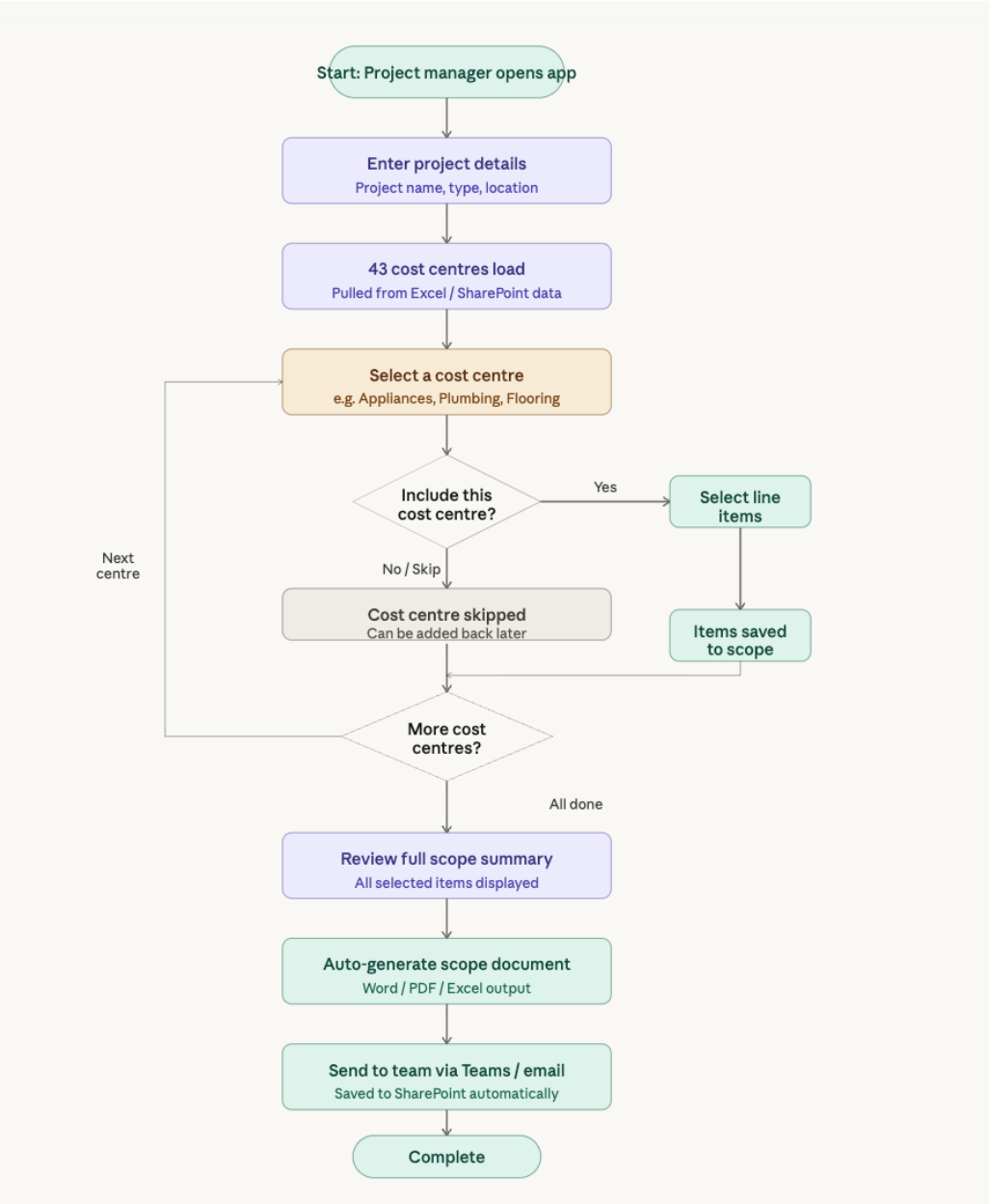
2.1 How it works today

- Project manager opens multiple Excel files (43 cost centre workbooks)
- Manually searches for relevant items in each cost centre tab
- Copies and pastes selected items into a scope document template
- Formats the document manually before sending to the team
- Team members then add pricing based on the scope provided

2.2 Key problems with the current approach

Problem	Impact
Time consuming	Each scope can take hours to compile manually
Error-prone	Items can be missed or incorrectly copied
Inconsistent	Different managers format scopes differently

No tracking	No record of who built which scope or when
Hard on tablets	Excel files are difficult to navigate on site



3. Solution Options

There are three possible approaches, ranked from most recommended to most complex. All three work within Microsoft 365.

Option A — Power Apps + Power Automate (RECOMMENDED)

Best for: Teams already using Microsoft 365. No extra cost. Works on tablets immediately.

Power Apps is Microsoft's built-in app builder. It connects directly to your Excel files stored in SharePoint and lets you build a point-and-click form that works on any tablet or phone. Power Automate then generates the scope document automatically when the form is submitted.

What you build:

1. A Power Apps canvas app with screens for project details, cost centre selection, and item selection
2. Excel files for each cost centre uploaded to SharePoint as the data source
3. A Power Automate flow that generates a Word scope document on submission
4. The completed scope saved back to SharePoint and emailed to the team

Pros:

- No extra software or licensing — included in Microsoft 365
- Works on iPad, Android tablet, or any browser
- Connects directly to your existing Excel data
- Microsoft provides free learning resources and tutorials
- Easy for the IT support person to maintain

Cons:

- Requires learning Power Apps (estimated 1-2 weeks to get comfortable)
- Complex logic can be harder to debug

Estimated build time:

- Weeks 1-2: Learn Power Apps basics + set up SharePoint data
- Weeks 3-4: Build the app screens
- Week 5: Build the Power Automate document generation flow
- Week 6: Testing and rollout

Option B — Microsoft Forms + Power Automate

Best for: Quickest start. Limited customisation but very easy to set up.

Microsoft Forms is even simpler than Power Apps. You build a form with dropdowns and checkboxes, and Power Automate collects the responses and generates a document. The limitation is that Forms does not dynamically filter items based on cost centre — all items show at once.

Pros:

- Extremely quick to set up (days, not weeks)
- Very simple to use on a tablet
- No learning curve — drag and drop form builder

Cons:

- Cannot dynamically filter items per cost centre (all 43 centres show at once)
- Limited formatting options for generated documents
- Not suitable if items change frequently

Option C — Custom Web Application (Advanced)

Best for: Maximum flexibility and best user experience. Requires coding skills.

Using Cursor (the AI coding tool you mentioned), you can build a full web application using React or Next.js. This gives you complete control over the design, can be opened on any tablet via a browser link, and can integrate with Microsoft 365 via the Graph API. This is the most powerful solution but requires the most effort.

Pros:

- Fully custom design — exactly what you want
- Can be made to look and feel like a professional app
- Works on any device via a browser link
- Can integrate with Microsoft Teams, SharePoint, Outlook

Cons:

- Requires JavaScript/React knowledge or significant use of Cursor AI assistance
- Needs hosting (Azure Static Web Apps — included free in many M365 plans)

- More complex to maintain
- Longer build time (6-10 weeks)

4. Recommended Solution — Detailed Design (Option A)

4.1 System Architecture Overview

The system has four layers that work together:

Layer	Tool Used	Purpose
Data Layer	SharePoint + Excel	Store all 43 cost centres and their items
App Layer	Power Apps Canvas App	Tablet-friendly selection interface
Automation Layer	Power Automate	Generate scope document automatically
Output Layer	Word / PDF / SharePoint	Final scope document stored and shared

4.2 Data Structure — How to organise your Excel files

You need one master Excel file in SharePoint with two sheets, or you can keep separate files per cost centre. The master approach is simpler to maintain.

Sheet 1 — Cost Centres (one row per cost centre)

CostCentreID	CostCentreName	Category	Active
CC001	Main Appliances	Interior	Yes
CC002	Plumbing	Services	Yes
CC003	Flooring	Interior	Yes
...	... (up to CC043)	...	...

Sheet 2 — Line Items (one row per item)

ItemID	CostCentreID	ItemName	Unit
IT001	CC001	Microwave — Standard	Each
IT002	CC001	Dishwasher — Built-in	Each
IT003	CC002	Toilet Suite	Each
IT004	CC002	Basin & Tap	Each

...	...	...	...

4.3 Power Apps — Screen-by-Screen Design

Screen 1: Project Details

- Text field: Project Name
- Dropdown: Project Type (e.g. Residential, Commercial, Renovation)
- Dropdown or text: Project Location
- Date picker: Scope Date
- Button: Next — goes to Screen 2

Screen 2: Cost Centre Selection

- Gallery showing all 43 cost centres as large, tap-friendly cards
- Each card shows: Cost Centre name, number of items available
- Toggle switch on each card: Include / Skip
- A 'Select All' button and 'Clear All' button for speed
- Progress bar showing how many have been reviewed
- Button: Review Selections — goes to Screen 3

Screen 3: Item Selection (per cost centre)

- Shows one cost centre at a time
- List of all items for that cost centre with checkboxes
- Quantity field next to each item
- Notes field for any special instructions
- Navigation: Previous Cost Centre / Next Cost Centre
- Skipped cost centres are greyed out and bypassed automatically

Screen 4: Review & Submit

- Full summary of all selected items grouped by cost centre
- Ability to go back and edit any section
- Button: Generate Scope Document
- Confirmation message with link to the generated file

5. Step-by-Step Implementation Guide

Follow these steps in order. Do not skip ahead. Each step builds on the last.

Phase 1 — Set Up Your Data (Week 1)

Step 1.1 — Create your SharePoint site

5. Go to sharepoint.com and sign in with your work account
6. Click '+ Create site' and choose 'Team site'
7. Name it: 'Scope Builder System'
8. Add your IT support person as a member

Step 1.2 — Upload your Excel data file

9. Open Excel and create a new workbook
10. Create Sheet 1 called 'CostCentres' with columns: CostCentreID, CostCentreName, Category, Active
11. Enter all 43 cost centres — one per row
12. Create Sheet 2 called 'LineItems' with columns: ItemID, CostCentreID, ItemName, Unit
13. Enter all line items for every cost centre
14. Save the file to your SharePoint site under Documents

Step 1.3 — Create a Word scope document template

15. Open Word and design your scope template
16. Add placeholder text that Power Automate will replace: <<ProjectName>>, <<ProjectType>>, <<ScopeDate>>
17. Add a section for each cost centre or use a table format
18. Save this template to SharePoint in a folder called 'Templates'

Phase 2 — Build the Power App (Weeks 2-4)

Step 2.1 — Open Power Apps

19. Go to make.powerapps.com
20. Sign in with your Microsoft 365 work account
21. Click '+ Create' then 'Canvas app from blank'
22. Name it 'Scope Builder' and choose 'Tablet' layout

Step 2.2 — Connect your data

23. In Power Apps, click the 'Data' icon on the left panel
24. Click 'Add data' and search for 'SharePoint'

25. Connect to your SharePoint site
26. Select your Excel file — Power Apps will show both sheets as data sources

Step 2.3 — Build Screen 1 (Project Details)

27. Add a Text Input for Project Name
28. Add a Dropdown and set Items to: ['Residential', 'Commercial', 'Renovation', 'Other']
29. Add another Dropdown for Project Location
30. Add a Date Picker
31. Add a Button labelled 'Next' with this formula in the OnSelect property:

Formula: Navigate(Screen2, ScreenTransition.Slide)

Step 2.4 — Build Screen 2 (Cost Centre Selection)

32. Add a Gallery control — choose 'Vertical gallery'
33. Set the Items property to your CostCentres table
34. Inside the gallery, add: a Label for CostCentreName, a Toggle control for Include/Skip
35. Add a 'Next' button that navigates to Screen 3

Step 2.5 — Build Screen 3 (Item Selection)

36. Add a variable to track which cost centre is being viewed
37. Add a Gallery filtered to show only items for the current cost centre
38. Add checkboxes next to each item
39. Add Previous and Next buttons to move through cost centres
40. Store selected items in a Collection (Power Apps memory)

Step 2.6 — Build Screen 4 (Review and Submit)

41. Add a Gallery showing all collected items grouped by cost centre
42. Add a 'Generate Scope' button
43. Set the OnSelect to trigger your Power Automate flow (built in Phase 3)

Phase 3 — Build the Automation (Week 5)

Step 3.1 — Create the Power Automate flow

44. Go to make.powerautomate.com
45. Click '+ Create' and choose 'Instant cloud flow'
46. Name it 'Generate Scope Document'
47. Set the trigger to 'PowerApps' (so it starts when the app calls it)

Step 3.2 — Add the document generation steps

48. Add action: 'Get file content' — select your Word template from SharePoint
49. Add action: 'Create file from template' — use the Word connector
50. Map the template placeholders to the data coming from Power Apps
51. Add action: 'Create file' in SharePoint to save the generated scope
52. Add action: 'Send an email' to notify the team with a link to the file

Phase 4 — Testing & Rollout (Week 6)

Step 4.1 — Internal testing

53. Run through the app yourself with a test project
54. Check that all 43 cost centres appear correctly
55. Verify the generated scope document matches what was selected
56. Test on an actual tablet — check buttons are large enough to tap

Step 4.2 — User testing

57. Ask one project manager to try it with a real project
58. Collect feedback on what is unclear or missing
59. Make adjustments based on feedback

Step 4.3 — Rollout

60. Share the app link with all project managers via Teams or email
61. Provide a simple 1-page user guide (how to open, how to use)
62. Set up a shared Teams channel for questions and feedback

6. Managing Cost Centres Over Time

6.1 Adding a new cost centre

Because the app reads from SharePoint Excel data, you never need to change the app itself when adding a cost centre. Simply:

63. Open the CostCentres sheet in your SharePoint Excel file
64. Add a new row with the new CostCentreID, name, and category
65. Add the line items to the LineItems sheet with the matching CostCentreID
66. The next time someone opens the app, the new cost centre will appear automatically

6.2 Disabling a cost centre temporarily

Set the 'Active' column to 'No' in the CostCentres sheet. The app can be configured to only show active cost centres, so it disappears from the list without deleting any data.

6.3 Changing items within a cost centre

Simply edit the LineItems sheet in SharePoint. Changes appear in the app immediately — no republishing needed.

7. Optional Future Enhancement — AI Agent Integration

You mentioned training an AI agent using the Excel files. This is a powerful future option once the core system is running. Here is how it would work:

7.1 Using Microsoft Copilot Studio

- Copilot Studio (formerly Power Virtual Agents) lets you build a custom AI chatbot connected to your SharePoint data
- You can upload your cost centre Excel files as knowledge sources
- Project managers could ask the agent natural language questions like: 'What items are in the Plumbing cost centre?' or 'Add all standard appliances to my scope'
- The agent can suggest cost centres based on project type automatically

7.2 Implementation approach

67. Wait until the core Power Apps system is stable and in use
68. Go to copilotstudio.microsoft.com
69. Create a new copilot and connect it to your SharePoint data

70. Train it with common questions and responses
71. Embed the copilot in your Power App as an optional assistant

> **Note:** This feature may require a Copilot Studio licence. Check your Microsoft 365 plan or speak with your IT support contact.

8. Learning Resources

Everything you need to learn Power Apps and Power Automate is available free from Microsoft. Use these resources in order:

Power Apps

Resource	Where to find it
Power Apps beginner course	learn.microsoft.com — search 'Power Apps canvas'
Power Apps YouTube channel	YouTube: search 'Microsoft Power Apps'
Power Apps community forum	powerusers.microsoft.com
Gallery + SharePoint tutorial	YouTube: 'Power Apps gallery SharePoint'

Power Automate

Resource	Where to find it
Power Automate beginner course	learn.microsoft.com — search 'Power Automate'
Word document generation flow	YouTube: 'Power Automate create Word document'
SharePoint integration guide	docs.microsoft.com — Power Automate SharePoint

9. Project Timeline Summary

Week	Phase	Key Tasks
Week 1	Data Setup	SharePoint site, Excel data file, Word template
Week 2	Learning	Complete Microsoft learn.microsoft.com Power Apps course
Week 3	Build App — Screens 1-2	Project details screen, cost centre gallery screen
Week 4	Build App — Screens 3-4	Item selection screen, review and submit screen
Week 5	Automation	Power Automate flow, document generation, email notification
Week 6	Testing & Rollout	Internal test, user test, feedback, deployment

Ongoing	Maintenance	Add cost centres via Excel, gather feedback, refine

10. Immediate Next Steps

Start with these three actions this week:

Action 1: Create your SharePoint site and upload your 43 cost centre Excel files.

Action 2: Complete the free Microsoft Power Apps beginner course at learn.microsoft.com

Action 3: Share this document with your IT support contact and walk through it together.

Once you have completed your Excel data setup, share it with your developer (or bring it back here) and the next step will be building Screen 1 of the Power App together.

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